



# Histograms

A HISTOGRAM IS A TYPE OF BAR GRAPH. IN A HISTOGRAM, THE AREA OF THE BARS, NOT THEIR LENGTH, REPRESENTS THE SIZE OF THE DATA.

## SEE ALSO

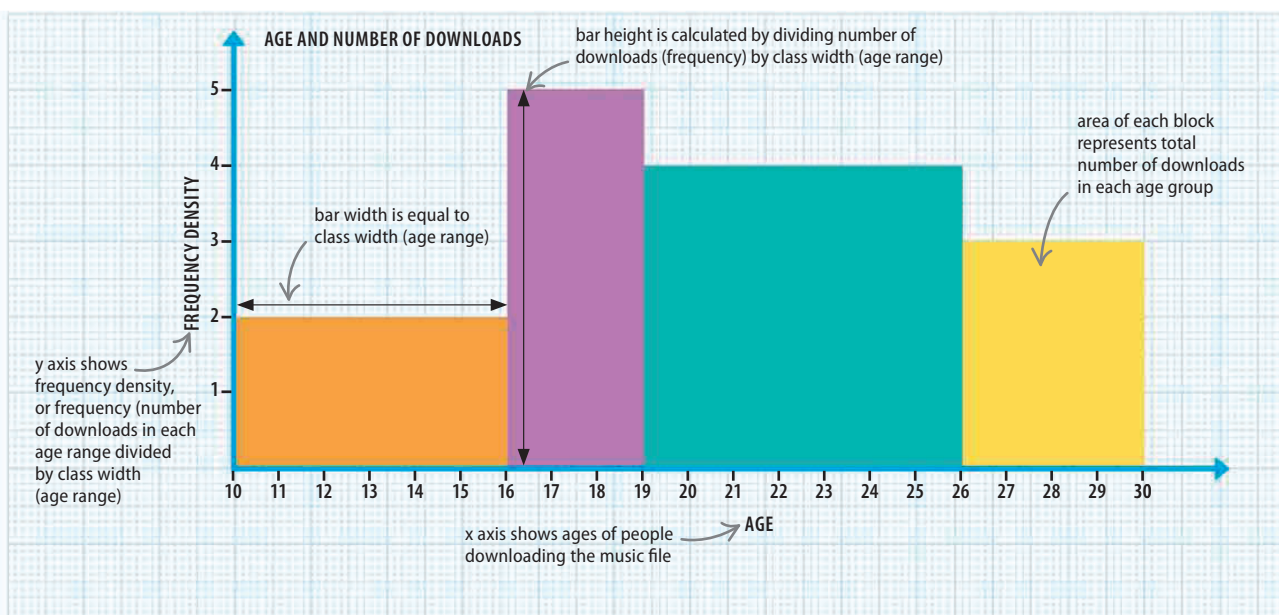
◀ 204–205 Collecting and organizing data

◀ 206–209 Bar graphs

◀ 220–223 Measuring spread

## What is a histogram?

A histogram is a diagram made up of blocks on a graph. Histograms are useful for showing data when it is grouped into groups of different sizes. This example looks at the number of downloads of a music file in a month (frequency) by different age groups. Each age group (class) is a different size because each covers a different age range. The width of each block represents the age range, known as class width. The height of each block represents frequency density, which is calculated by dividing the number of downloads (frequency) in each age group (class) by the class width (age range).



## LOOKING CLOSER

### Histograms and bar graphs

Bar graphs look like histograms, but show data in a different way. In a bar graph, the bars are all the same width. The height of each bar represents the total (frequency) for each group, while in a histogram, totals are represented by the area of the blocks.

#### ► Bar graph

This bar graph shows the same data as shown above. Although class widths are different, the bar widths are the same.

